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2024 Higher School Certificate Trial Examination Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- NESA approved calculators may be used
- Write your student number on the paper and the multiple-choice answer sheet
- A reference sheet is provided at the back of this paper
- In Questions 11-32, show relevant mathematical reasoning and/or calculations
- Diagrams not to scale

Total marks:
100

Section I 10 marks

- Attempt Questions 1-10 (pages 2-6)
- Allow about 15 minutes for this section

Section II 90 marks

- Attempt Questions 11-32 (pages 7-27)
- Allow about 2 hours and 45 minutes for this section

Disclaimer

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2024-MAA-NSW-Y12-NA-EX-QATS

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet provided for Questions 1-10

- 1 A game is played where a coin is flipped, and a card is drawn at random from a standard 52 card deck. If the card is hearts and the coin shows a head the player wins, otherwise the player loses.

What is the probability that a player wins?

- A. 0.125
- B. 0.25
- C. 0.5
- D. 0.725

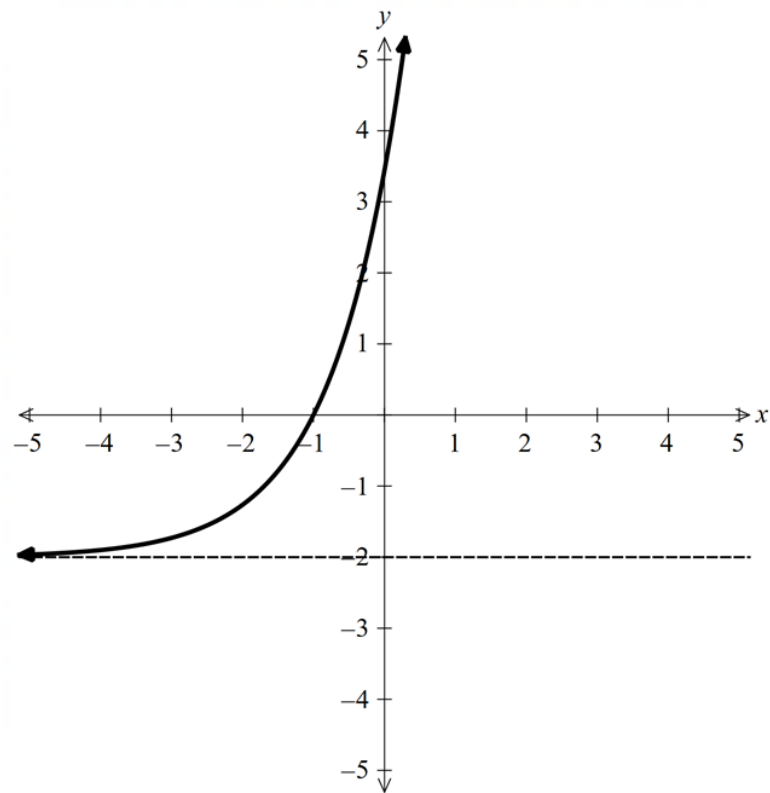
- 2 What is the sum of the first 20 terms of the arithmetic sequence -23, -19, -15, ...?

- A. 300
- B. 600
- C. 1220
- D. 1260

- 3 What is the domain of $f(x) = \sqrt{4x - 8}$?

- A. $x \geq -8$
- B. $x \geq 2$
- C. $x \geq 4$
- D. $x \geq 8$

- 4 The graph of $y = 2e^{x+b} + c$ is given below.



What is the value of b ?

- A. -2
- B. -1
- C. 1
- D. 2

- 5 In an assessment task, students received a mark out of 30. The set of marks had a mean of 17 and standard deviation of 3. The marks were then standardised to a normal distribution with mean of 70% and standard deviation of 5%.

A student received a mark of 23 in the task.

What is their standardised mark?

- A. 23%
- B. 55%
- C. 75%
- D. 80%

- 6 A random variable is defined by the probability distribution:

$$P(X = x) = \frac{x^2 + x}{20} \text{ for } x = 1, 2, 3.$$

What is the value of $Var(X)$?

- A. 0.20
- B. 0.45
- C. 0.67
- D. 2.50

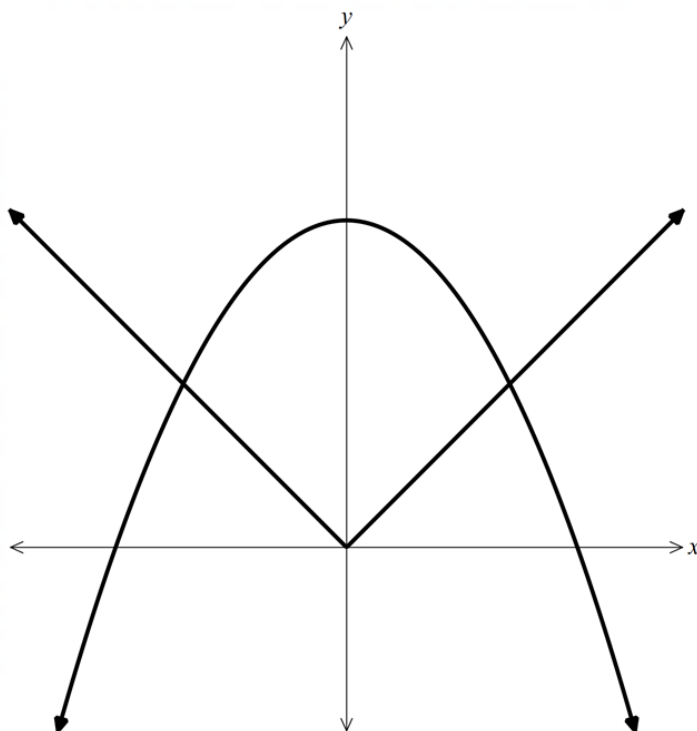
- 7 For two functions $g(x)$ and $h(x)$, it is known that for $a > 0$ that $g(a) > h(a)$.

It is also known that $h'(x) < 0$ and $g'(x) > 0$ for $a > 0$.

How many solutions are there to $g(x) - h(x) = 0$ in the domain $a > 0$?

- A. 0
- B. 1
- C. 2
- D. 3

- 8 The diagram below shows the curves $y = |x|$ and $y = 2 - x^2$.



What is the area between the two curves?

- A. $\frac{7}{6}$ units²
- B. $\frac{14}{3}$ units²
- C. $\frac{7}{3}$ units²
- D. $\frac{13}{3}$ units²

- 9** How many solutions are there to the equation $x - 2\pi \cos x = 0$ in the domain $0 \leq x \leq 2\pi$?

- A. 0
- B. 1
- C. 2
- D. 3

- 10** Consider the sequence,

$$\log_2 3, \log_2 9, \log_2 27 \dots$$

What is the sum of the first 10 terms of this sequence?

- A. $3 \log_2 3$
- B. $10 \log_2 3$
- C. $30 \log_2 3$
- D. $55 \log_2 3$

Section II

90 marks

Attempt Questions 11-32

Allow about 2 hours and 45 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

In Questions 11-32 your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (2 marks)

Solve $\log_x 125 = 3$. **2**

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Question 12 (3 marks)

Given that $f(x) = xe^x$, find $f'(1)$. **3**

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Question 13 (2 marks)

In a particular board game, a six-sided die is rolled to determine how a player moves their game piece around the board.

If the number rolled is odd, the player moves their game piece forward the number of spaces shown on the die.

If the number rolled is even, the player does not move.

Calculate the expected number of spaces moved each turn. 2

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Question 14 (2 marks)

Evaluate $\int_{-2}^1 3x^3 dx$ 2

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Question 15 (3 marks)

Find the equation of the normal, in general form, to the curve $y = (2 - x)^4$ at the point $(3, 1)$.

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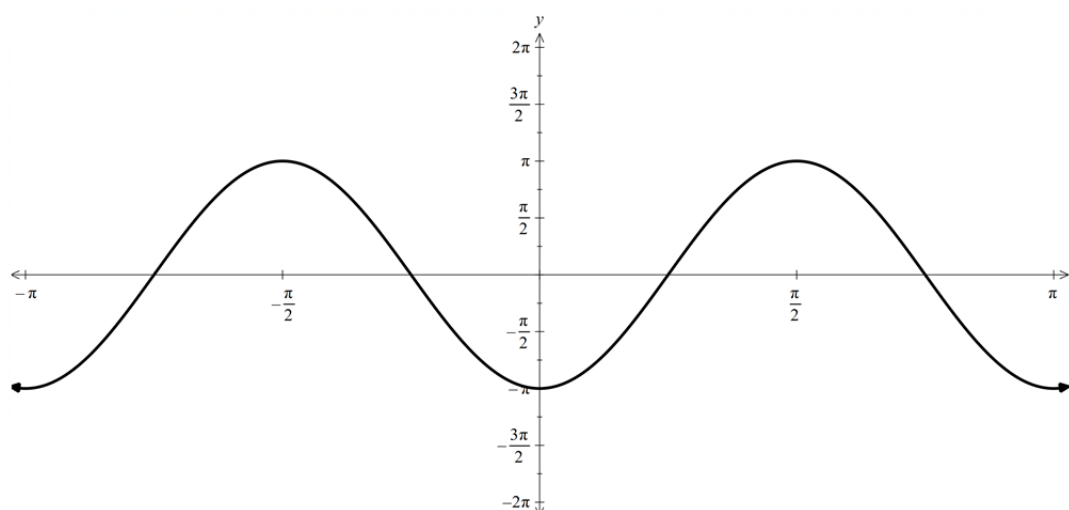
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Question 16 (3 marks)

The graph of $y = k\cos(a(x + b))$ is shown.



What are the values of a , b and k ?

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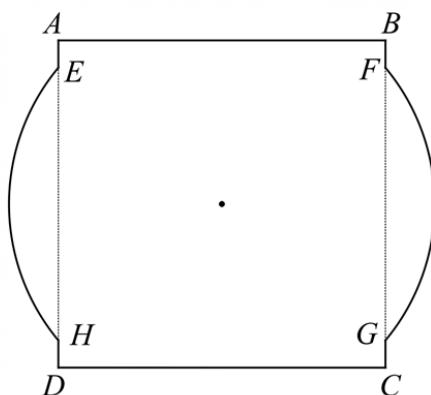
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Question 17 (4 marks)

The diagram shows a shape that consists of a square with two identical arcs on opposite sides of the square.



The square has a side length of 12m and the arcs have their centre at the centre of the square.

The points E, F, G and H are the endpoints of the arcs.

$$AE = FB = CG = HD = 1 \text{ m.}$$

Calculate the area of the shape, to the nearest m^2 .

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Question 18 (4 marks)

The table below shows the contribution per period for an annuity with a future value of \$1.

Number of Periods	Interest rate (% per period)			
	0.25%	0.5%	0.75%	1%
1	1.0000	1.0000	1.0000	1.0000
3	0.3325	0.3317	0.3308	0.3300
6	0.1656	0.1646	0.1636	0.1625
9	0.1100	0.1089	0.1078	0.1067
12	0.0822	0.0811	0.0800	0.0788
15	0.0655	0.0644	0.0632	0.0621

- (a) Arthur is saving for a holiday. He is able to contribute a fixed amount, \$Q, each quarter to an annuity fund with an interest rate of 4% p.a.

He plans to do this for 3 years and hopes to have \$8000 in the account at the end of the 3 years.

Determine the quarterly contribution \$Q and the amount of interest earned.

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- (b) Two years into the investment, Arthur's friend Aimee decides she will join him on the holiday. She invests \$400 per month into an account that earns 6% p.a. for 1 year.

How much will Aimee have in her account after 1 year?

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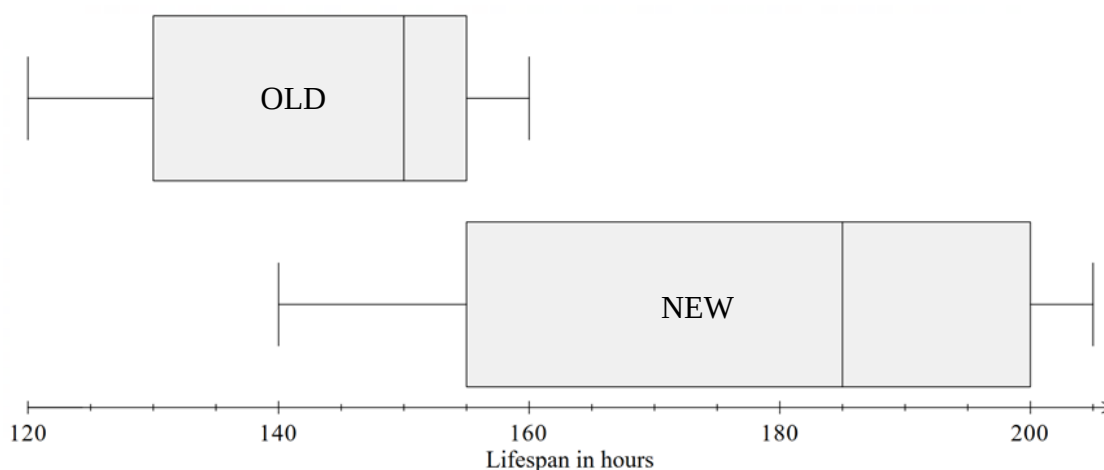
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Question 19 (5 marks)

A manufacturer of light globes claims that their new globe has a lifespan that is 30% longer than their old globes, and that their new globes are more consistent throughout their lifespan.

The box plots below show data for a sample of old and new globes.



- (a) State the median for each sample of globes. **1**

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- (b) Describe the shape of the distribution for the old globes. **1**

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- (c) Comment on the manufacturer's claims about their new globes. **3**

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Question 20 (4 marks)

- (a) Differentiate $\sqrt{3 + x^3}$. 2

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- (b) Hence, find $\int \frac{x^2}{\sqrt{3 + x^3}} dx$. 2

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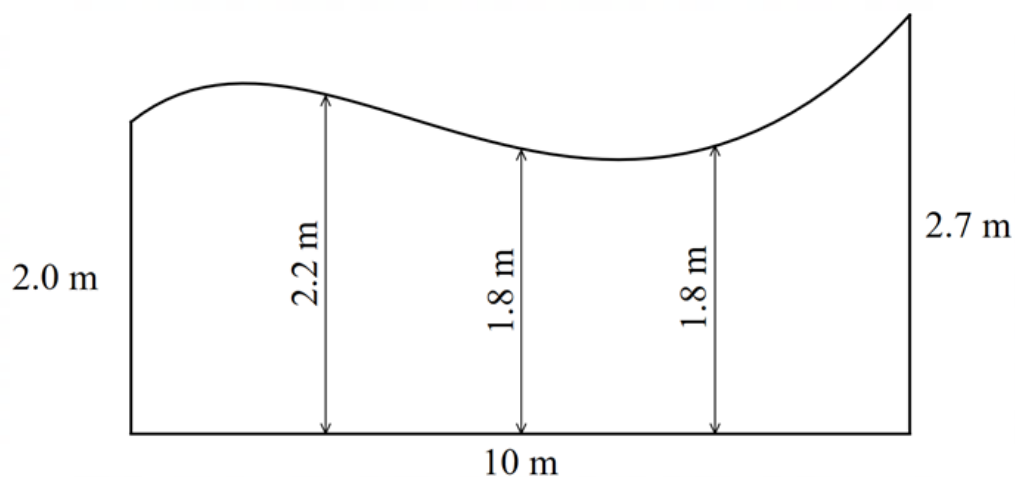
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Question 21 (3 marks)

A wall has been built with a curved top as shown.



Use the trapezoidal rule, along with all the available dimensions, to calculate the area of the face of the wall.

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Question 22 (6 marks)

During her ALF training, Lucy takes 20 shots at goal from a range of different distances.

She records the percentage of goals she kicked for each distance.

Distance (m)	10	15	20	25	30	35	40
Goals kicked (%)	100	95	85	80	70	50	45

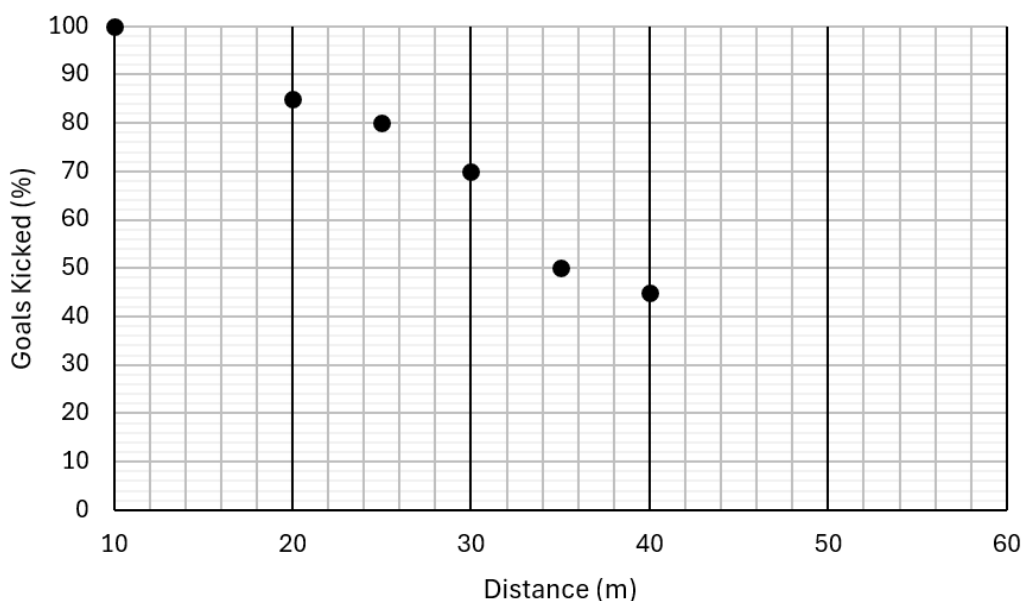
- (a) Calculate Pearson's correlation coefficient and hence, describe the correlation between the percentage of goals kicked and the distance from the goal. 2

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- (b) The line of best fit will pass through the point $(\bar{x}, \bar{y})$.

Mark this point on the graph below and hence draw a line of best fit. 2



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Question 22 continues on the next page

Question 22 (continued)

- (c) Use your line of best fit to predict the percentage of goals kicked at a distance of 50 m. 1

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- (d) Lucy can kick the ball a maximum distance of 45 m. Discuss the validity of the model in making predictions beyond a distance of 45 m. 1

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Question 23 (3 marks)

A company manufactures cans of soft drink. The volume inside each can is normally distributed with a mean of 375 mL and a standard deviation of 3 mL. It is also known that approximately 99% of the cans have a volume of less than 382 mL.

During the quality control process, cans are rejected if they are underfilled or overfilled by a certain amount.

1% of the cans are rejected for being underfilled and 2.5% of cans are rejected for being overfilled.

Between what two volumes do the cans that are not rejected lie? 3

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Question 24 (5 marks)

An arithmetic and geometric sequence both have first term 1, and their second terms are equal.

The 14th term of the arithmetic sequence is three times the third term of the geometric sequence.

Given that both sequences contain only integers, find the 20th term of each sequence. 5

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Question 25 (3 marks)

When Lucio plays for his soccer team, they have an 85% chance of winning. When he does not play, their chance of winning is 40%.

The chance that Lucio plays any particular game is 90%.

- (a) Find the probability that Lucio plays and his team wins. 1

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- (b) Find the probability that Lucio played, given that his team won. 2

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Question 26 (4 marks)

A continuous random variable X is defined by the probability density function:

$$f(x) = 6 \cos 3x \text{ for } 0 \leq x \leq a.$$

- (a) Find the exact value of a . 3

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- (b) Find the mode of X . 1

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Question 27 (5 marks)

A substance decomposes at a rate that is proportional to the mass of the substance remaining.

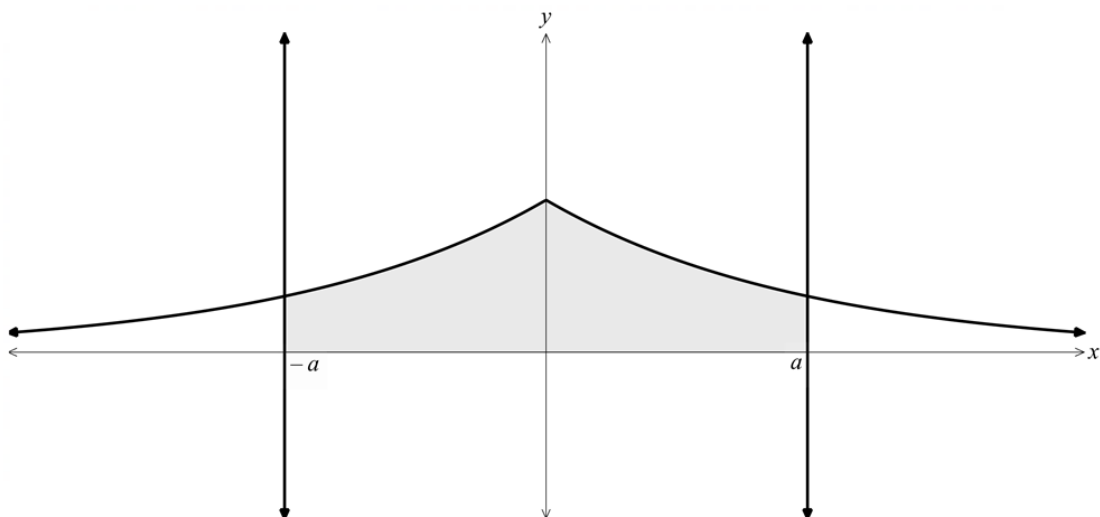
If initially there is 15 kg of the substance and after 20 hours half of the substance has decomposed, determine how long it will take for 90% of the substance to decompose. (Give your answer to the nearest hour.)

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Question 28 (4 marks)

The graph below shows the area bounded by the curve $y = e^{-|x|}$, the x -axis and the lines $x = \pm a$.



- (a) Show that the area is $2(1 - e^{-a})$ units².

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- (b) Explain what happens to the area as $a \rightarrow \pm\infty$.

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Question 29 (6 marks)

A particle moves in a straight line so that its velocity, in ms^{-1} , is given by:

$$v = \frac{4t}{1+t^2}$$

The particle is initially at rest at the origin.

- (a) Determine the displacement equation for the particle. 2

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- (b) Determine when the particle reaches its maximum velocity and state the exact location of the particle when this occurs. 4

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Question 30 (6 marks)

Erin has an investment account into which she deposits \$2000 at the beginning of each month. The account earns interest at 6% p.a. compounded monthly. After the interest has been calculated at the end of the month, she then withdraws \$ M to cover her expenses for the next month.

- (a) Show that the amount in the account after n months is given by:

$$A_n = (402000 - 200M)(1.005^n - 1) \quad 4$$

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Question 30 continues on the next page

Question 30 (continued)

- (b) After 30 months, Erin has \$10 000 in the account. Find the value of $\$M$, correct to the nearest dollar.

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[illegible]

Question 31 (4 marks)

Sketch the graph of $y = 1 - e^{-x^2}$, showing any turning points and asymptotes.

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Question 31 continues on next page

Question 31 (continued)

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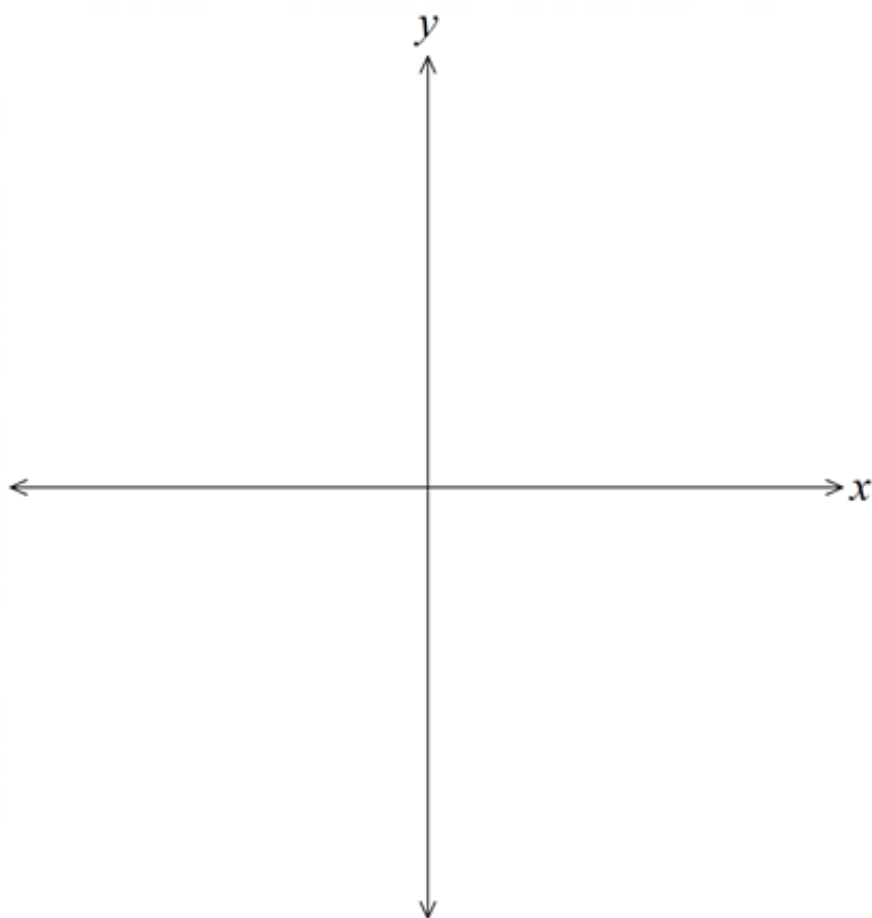
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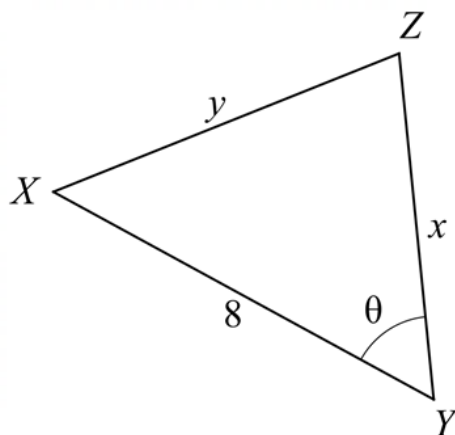
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Question 32 (9 marks)

Triangle XYZ has a perimeter of 20 cm.



(a) Show that $\cos \theta = \frac{3x - 10}{2x}$.

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Question 32 continues on the next page

Question 32 (continued)

- (b) If the area of the triangle is A , show that $A^2 = -20(x^2 - 12x + 20)$. 3

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- (c) Hence, determine the maximum area of the triangle, and comment on the triangle's shape when the area is a maximum. 4

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Question 32 continues on next page

[illegible]

END OF EXAMINATION

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

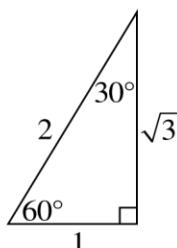
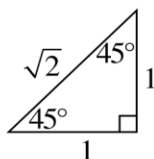
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1 + t^2}$$

$$\cos A = \frac{1 - t^2}{1 + t^2}$$

$$\tan A = \frac{2t}{1 - t^2}$$

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

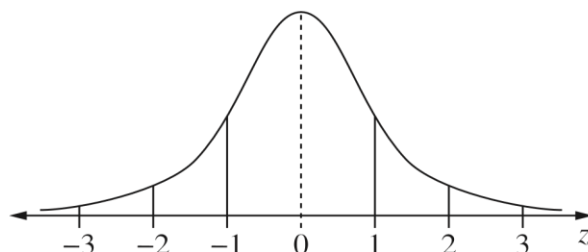
$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq x) = \int_a^x f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1 - p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1 - p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})]\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z &= a + ib = r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

Student Number:.....

Mathematics Advanced – Multiple-Choice Answer Sheet

ATTEMPT ALL QUESTIONS

Place an X in your response for each multiple-choice question to indicate your choice.

Question				
1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D